

Linux User and Group Management

- Task: Create users and set passwords.
- Create groups and assign users.
- Setup shared directories with group permission.

Sudo `useradd -m tokyo -s /usr/bin/bash`: this command is used to add user to the home directory using '-m', '-s' flag used to set the path to the user's login shell.

Sudo `passwd tokyo`: command is used to set password to the user.

```
ubuntu@ip-172-31-42-193:/home$ sudo useradd -m tokyo -s /usr/bin/bash -p test@123
ubuntu@ip-172-31-42-193:/home$ sudo passwd tokyo
New password:
Retype new password:
passwd: password updated successfully

ubuntu@ip-172-31-42-193:/home$ sudo useradd -m berlin -s /usr/bin/bash
ubuntu@ip-172-31-42-193:/home$ sudo passwd berlin
New password:
Retype new password:
passwd: password updated successfully

ubuntu@ip-172-31-42-193:~$ which bash
/usr/bin/bash
ubuntu@ip-172-31-42-193:~$ sudo useradd -m professor -s /usr/bin/bash
ubuntu@ip-172-31-42-193:~$ sudo passwd professor
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-42-193:~$
```

Cat `/etc/passwd`: This file is world-readable and lists all system and regular user accounts, including system services that run as specific users.

```
tokyo:x:1001:1001::/home/tokyo:/usr/bin/bash
berlin:x:1002:1002::/home/berlin:/usr/bin/bash
professor:x:1003:1003::/home/professor:/usr/bin/bash
```

Cd `/home`: command is used to change the directory to home. To determine whether the user is created or not.

```
ubuntu@ip-172-31-42-193:~$ cd /home
ubuntu@ip-172-31-42-193:/home$ ls
berlin professor tokyo ubuntu
ubuntu@ip-172-31-42-193:/home$
```

- Task 2: Create Group
- Developers
- Admins
- Verify check /etc/group

Sudo groupadd developers: command is used to create new group.

```
ubuntu@ip-172-31-42-193:~$ sudo groupadd developers
ubuntu@ip-172-31-42-193:~$ sudo groupadd admins
ubuntu@ip-172-31-42-193:~$ cat /etc/group
```

Cat /etc/group: This command is used to show all the groups on the system, including which users belong to which group. Groups are used to manage file permission and access control for multiple users.

```
ubuntu:x:1000:
tokyo:x:1001:
berlin:x:1002:
professor:x:1003:
developers:x:1004:
admins:x:1005:
```

- Task 3: Assign to groups
- Assign users:
- Tokyo – developers
- Berlin – developers + admins
- Professor – admins
- Verify the user's appropriate command to check out group membership.

Sudo gpasswd -a tokyo developers

Sudo gpasswd -a berlin developers

Sudo gpasswd -a berlin admins

Sudo gpasswd -a professor admins

This command is used to administer /etc/group. Every group can have administrators, members, and a password.'-a' flag used to add user to the named group.

```
ubuntu@ip-172-31-42-193:~$ sudo gpasswd -a tokyo developers
Adding user tokyo to group developers
ubuntu@ip-172-31-42-193:~$ sudo gpasswd -a berlin developers
Adding user berlin to group developers
ubuntu@ip-172-31-42-193:~$ sudo gpasswd -a berlin admins
Adding user berlin to group admins
ubuntu@ip-172-31-42-193:~$ sudo gpasswd -a professor admins
Adding user professor to group admins
```

Cat /etc/group: This command is used to show all the groups on the system, including which users belong to which group. Groups are used to manage file permission and access control for multiple users.

```
ubuntu:x:1000:
tokyo:x:1001:
berlin:x:1002:
professor:x:1003:
developers:x:1004:tokyo,berlin
admins:x:1005:berlin,professor
ubuntu@ip-172-31-42-193:~$ █
```

- Task 4; shared Directories
- Set permission to 775
- Set group owner to developers
- Test by creating files as tokyo an berlin
- Verify check permission and test file creation.

Sudo mkdir /opt/dev-project: This command creates a directory; sudo gives administrative access, allowing the creation of directories in system protected locations. This is used to organize project files in a centralized location.

```
ubuntu@ip-172-31-42-193:~$ sudo mkdir /opt/dev-project
ubuntu@ip-172-31-42-193:~$ █
```

sudo chgrp developer /opt/dev/project: this command is used to change group ownership.

```
ubuntu@ip-172-31-42-193:~$ sudo chgrp developers /opt/dev-project
ubuntu@ip-172-31-42-193:~$
```

```
ubuntu@ip-172-31-42-193:~$ ls -la /opt/dev-project
total 8
drwxr-xr-x 2 root developers 4096 Feb  8 11:52 .
drwxr-xr-x 3 root root      4096 Feb  8 11:52 ..
ubuntu@ip-172-31-42-193:~$
```

sudo chmod 775 /opt/dev-project: this command used to change file mode bits. Which gives permission to the read, write, execute permission to owner, and same permission to group, for others it is only read and execute.

```
ubuntu@ip-172-31-42-193:~$ sudo chmod 775 /opt/dev-project
ubuntu@ip-172-31-42-193:~$
```

Su berlin: this command is used to switch the user

Cd /opt/dev-project echo "Hi i am berlin!" > intro.txt: The command creates file in the path which is in root but added in the same group in which user is present the so that user can also access the file.

Su tokyo: this command is used to switch the user

Cd /opt/dev-project echo "Hi i am tokyo" > tokyo_intro.txt: The command creates file in the path which is in root but added in the same group in which user is present the so that user can also access the file.

```

tokyo@ip-172-31-42-193:/opt/dev-project$ echo "Hi i am tokyo!" > tokyo_intro.txt
tokyo@ip-172-31-42-193:/opt/dev-project$ ls
intro.txt  tokyo_intro.txt
tokyo@ip-172-31-42-193:/opt/dev-project$ exit
exit
ubuntu@ip-172-31-42-193:/home$ su berlin
Password:
berlin@ip-172-31-42-193:/home$ cd
berlin@ip-172-31-42-193:~$ cd /opt/dev-project
berlin@ip-172-31-42-193:/opt/dev-project$ ls
intro.txt  tokyo_intro.txt
berlin@ip-172-31-42-193:/opt/dev-project$ █

berlin@ip-172-31-42-193:/opt/dev-project$ echo "Hi i am berlin!" > intro.txt
berlin@ip-172-31-42-193:/opt/dev-project$ exit
exit
ubuntu@ip-172-31-42-193:~$ su tokyo
Password:
tokyo@ip-172-31-42-193:/home/ubuntu$ cd
tokyo@ip-172-31-42-193:~$ cd /opt/dev-project/
tokyo@ip-172-31-42-193:/opt/dev-project$ ls
demo.txt  intro.txt
tokyo@ip-172-31-42-193:/opt/dev-project$ █

```

- Task 5: Team workspace
- Create user nairobi with home directory
- Create group project team
- Add nairobi and tokyo to project-team
- Create /opt/team-workspace directory
- Test by creating files at nairobi

Sudo useradd -m nairobi -s /usr/bin/bash: this command is used to add user to the home directory using '-m', '-s' flag used to set the path to the user's login shell.

Cd /home ls: command is used to set password to the user.

```

ubuntu@ip-172-31-42-193:~$ sudo useradd -m nairobi -s /usr/bin/bash
ubuntu@ip-172-31-42-193:~$ cd /home
ubuntu@ip-172-31-42-193:/home$ ls
berlin  nairobi  professor  tokyo  ubuntu
ubuntu@ip-172-31-42-193:/home$ sudo passwd nairobi
New password:
Retype new password:
passwd: password updated successfully

```

Sudo gpasswd -a nairobi project-team

Sudo gpasswd -a tokyo project-team

This command is used to administer /etc/group. Every group can have administrators, members, and a password.’-a’ flag used to add user to the named group.

```
ubuntu@ip-172-31-42-193:/home$ sudo gpasswd -a nairobi project-team
Adding user nairobi to group project-team
ubuntu@ip-172-31-42-193:/home$ sudo gpasswd -a tokyo project-team
Adding user tokyo to group project-team
nairobi:x:1006:
project-team:x:1007:nairobi,tokyo
ubuntu@ip-172-31-42-193:/home$
```

Sudo mkdir /opt/team-workspace: This command creates a directory; sudo gives administrative access, allowing the creation of directories in system protected locations. This is used to organize project files in a centralized location.

```
ubuntu@ip-172-31-42-193:/home$ sudo mkdir /opt/team-workspace
ubuntu@ip-172-31-42-193:/home$
```

Sudo chgrp project-team /opt/team-workspace: this command is used to change group ownership.

Sudo chmod 775 /opt/team-workspace: this command used to change file mode bits. Which gives permission to the read, write, execute permission to owner, and same permission to group, for others it is only read and execute.

Ls –la /opt/team-workspace: this command is used to list the directory contents ‘-a’ don’t ignore entries start with, ‘-l’ use a long listing format.

```
ubuntu@ip-172-31-42-193:/home$ sudo mkdir /opt/team-workspace
ubuntu@ip-172-31-42-193:/home$ sudo chgrp project-team /opt/team-workspace
ubuntu@ip-172-31-42-193:/home$ sudo chmod 775 /opt/team-workspace
ubuntu@ip-172-31-42-193:/home$ ls -la /opt/team-workspace/
total 8
drwxrwxr-x 2 root project-team 4096 Feb  9 06:57 .
drwxr-xr-x 4 root root          4096 Feb  9 06:57 ..
ubuntu@ip-172-31-42-193:/home$
```

Su nairobi: This command used to switch the user.